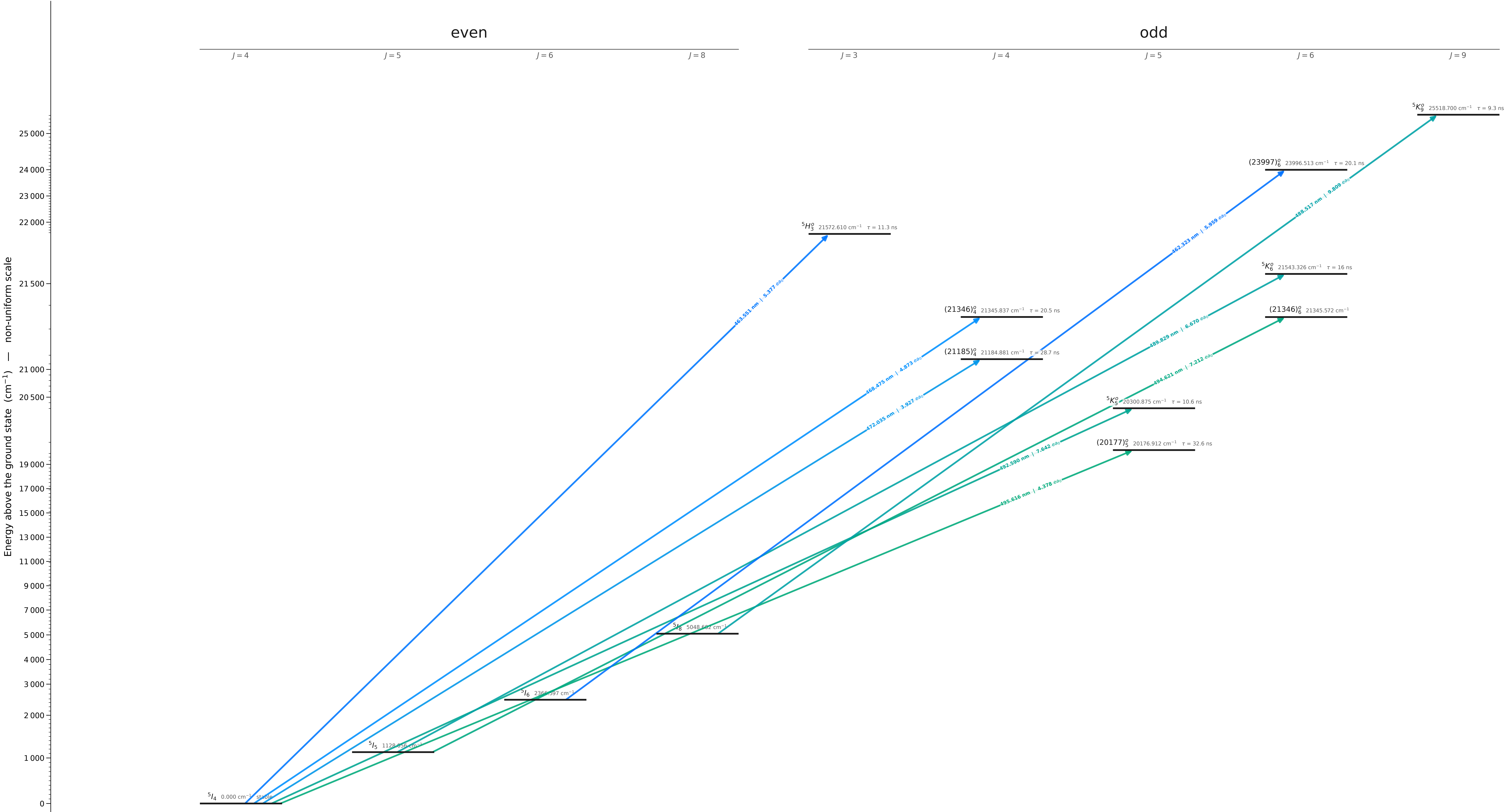
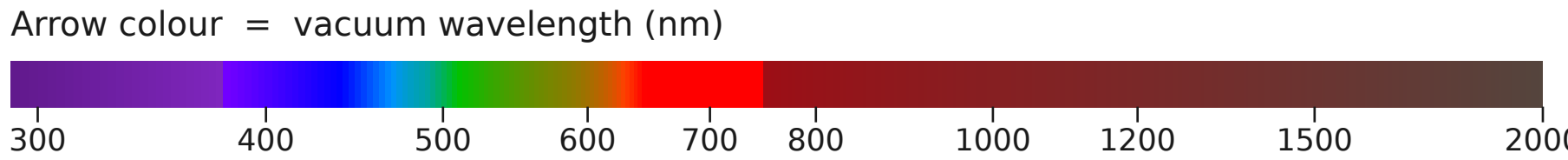


Neodymium

Nd (Nd I). Strongest classified transitions below 2 μm (NIST): 13 levels, 9 lines.

Nd⁺ ionisation limit 44560.11 cm⁻¹ = 5.52475 eV (224.416 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J || e r || J' \rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	(J er J') (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁵ I ₆ - (²³⁹⁹⁷) ₆ ^o	462.3226	462.1932	648.4486	5.959	5.600e+07	7.918	100	
⁵ I ₄ - ⁵ H ₃ ^o	463.5508	463.4210	646.7306	5.377	8.400e+07	14.08	94.9	
⁵ I ₄ - (²¹³⁴⁶) ₆ ^o	468.4754	468.3443	639.9321	4.873	5.200e+07	7.764	100	
⁵ I ₄ - (²¹¹⁸⁵) ₈ ^o	472.0347	471.9027	635.1068	3.927	3.300e+07	5.545	94.7	
⁵ I ₈ - ⁵ K ₈ ^o	488.5174	488.3810	613.6781	9.809	8.800e+07	17.11	81.8	
⁵ I ₅ - ⁵ K ₆ ^o	489.8294	489.6927	612.0344	6.670	5.900e+07	9.947	94.4	
⁵ I ₄ - ⁵ K ₅ ^o	492.5896	492.4521	608.6049	7.642	9.000e+07	15.01	95.4	
⁵ I ₅ - (²¹³⁴⁶) ₆ ^o	494.6206	494.4826	606.1059	7.212	6.700e+07	-	-	
⁵ I ₄ - (²⁰¹⁷⁷) ₅ ^o	495.6160	495.4777	604.8886	4.378	2.900e+07	4.882	94.5	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁴³Nd hyperfine structure (*I* = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)						
⁵ I ₄	-195.649(9)	122.25(28)	F=1/2: +3483.89	F=3/2: +3174.98	F=5/2: +2663.25	F=7/2: +1953.37	F=9/2: +1051.90	F=11/2: -32.75	F=13/2: -1290.27 F=15/2: -2708.52

Measured A, B constants; sources in the data file.